

Proprietary Data × Checkable Outcomes: The Shared Template Behind AI's Largest Vertical Wins, and a Forward Filter for Building Defensible Data Assets

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Abstract

AI's most celebrated vertical breakthroughs are usually narrated as isolated triumphs of architecture or scale. This paper argues they instead share a single, transferable template: a proprietary or hard-to-assemble dataset paired with an objective, machine-checkable outcome. AlphaFold2 reached a median 92.4 GDT_TS at CASP14 because the Protein Data Bank supplied experimentally-resolved ground truth and GDT supplied an automatic metric [1] [2]. SWE-bench made code progress measurable because real GitHub fixes are verified by unit tests — best system 1.96% at launch [4], a public-data benchmark whose capability has since diffused across labs (frontier SWE-bench Verified now 80.9% [21]) even as contamination-resistant successors keep the genuine frontier low (best model 17.8% on SWE-Bench Pro's proprietary commercial set [17]). FrontierMath kept math an open frontier because answers auto-verify yet remain <2% solved [5]. BloombergGPT's edge traced to a 363B-token proprietary corpus [6]. We formalize the template into a two-axis selection filter (data-propietarinness × outcome-verifiability), validate it against benchmark trajectories, and connect it to the 2025-2026 capital surge into verifiable reinforcement-learning environments — Anthropic reportedly discussing >\$1B in annual spend, with exclusive environment deals priced 4-5× non-exclusive [7][8], and recent work showing that scaling the breadth of verifiable environments outperforms pouring more compute into a fixed one [19]. We provide a methodology for scoring candidate verticals, supply benchmark tables with verbatim sourced numbers, and surface the limitations: the filter scores data shape only, public-data verticals (code, math) score lower on defensibility, and willingness-to-pay must be validated independently.

1. Introduction

The dominant public narrative around AI's vertical successes is one of architecture and scale: AlphaFold 'solved' protein folding with a clever attention mechanism; coding agents got better because models got bigger; finance LLMs work because Bloomberg trained a big model. Each story is told in isolation, in its own field's vocabulary, and credited to its own technical novelty [1][3][4][6].

This paper advances a different reading. When these wins are placed side by side and stripped of domain-specific framing, a single structural pattern recurs in every one: each pairs a proprietary or hard-to-assemble dataset with an objective, machine-checkable outcome. Protein structure had decades of experimentally-resolved structures plus the GDT metric [1]. Code has executable unit tests [4]. Competition and research mathematics have answers that auto-verify [5]. Finance had a 363-billion-token proprietary corpus that competitors could not assemble [6].

The synthesis is non-obvious precisely because these are celebrated results in unrelated fields, never analyzed together as one class. Extracting the shared template — proprietary data × verifiable outcome — converts a set of

post-hoc success stories into a forward-looking filter for choosing which vertical to build a data asset in next. That filter is the contribution of this paper.

We make this concrete. Section 3 formalizes the two-axis filter and a scoring methodology. Sections 4-5 validate it against benchmark data with verbatim-sourced numbers and against the 2025-2026 market for reinforcement-learning environments, where capital is now flowing toward exactly the proprietary-plus-verifiable structure the template predicts. Section 8 states the limitations sharply: the filter scores data shape, not business viability, and high verifiability without proprietary data (code, math) yields a thin moat.

2. Background and Related Work

Two literatures bracket this paper. The first is the data-exhaustion literature. Villalobos et al. estimate the effective stock of quality- and repetition-adjusted human-generated public text at roughly 300 trillion tokens and project, with 80% confidence, that language models will fully utilize it between 2026 and 2032 — earlier under heavy overtraining [9]. This 'data wall' makes the question of where fresh, defensible training signal comes from economically urgent rather than academic.

The second is the verifiable-reward literature. Reinforcement Learning with Verifiable Rewards (RLVR) was named and scoped in Tulu 3 explicitly to 'tasks with verifiable outcomes such as mathematics and verifiable instruction following,' replacing a learned reward model with a deterministic verification function [10]. DeepSeek-R1-Zero showed that large-scale RL on verifiable signal alone, with no supervised fine-tuning, produces emergent reasoning (e.g. AIME 2024 pass@1 rising from 15.6% to 71.0%) [11]. Process supervision, where each reasoning step is checked, outperforms outcome supervision and solves 78% of a representative MATH test subset [12]. More recent work scales the supply side of this signal directly: RLVE shows that growing a collection of adaptive verifiable environments (RLVE-Gym, 400 environments) lifts a 1.5B reasoning model by +3.37% across six benchmarks, versus only +0.49% from spending 3x more compute on a fixed environment [19]. The common thread is that a machine-checkable outcome is what makes the signal trainable at all — and that the scarce, scalable asset is the breadth of verifiable environments themselves.

Our contribution sits between these literatures. The data-wall work says human text is running out; the RLVR work says verifiable environments produce fresh signal. Neither asks the vertical-selection question this paper poses: given that verifiable, proprietary data is the scarce asset, which verticals actually possess both properties? We answer by mining the historical record of vertical wins for the recurring structural template, then turning that template into a filter.

Adjacent vertical-LLM results inform the data axis. BioBERT-style in-domain pretraining beats general-model adaptation in biomedicine [paper 2007.15779, related insight i11]; Med-PaLM 2 reached 86.5% on MedQA, a ~19-point gain over its predecessor's 67.6%, on the strength of clinical fine-tuning data [13]; MEDITRON-70B closed to within 5% of GPT-4 via a curated medical corpus [14]. These reinforce that, within a vertical, the corpus — not the method — is the durable variable.

3. Methodology: The Two-Axis Filter and Scoring Protocol

We define two orthogonal axes. Axis A, data-proprietaryness, measures how hard the underlying dataset is for a competitor to assemble: scored low when the data is on the open web (public GitHub, arXiv), high when it requires privileged access, exclusive licensing, expensive wet-lab generation, or accumulated operational history. Axis B, outcome-verifiability, measures whether task success can be checked automatically and objectively without a human judge: scored low for open-ended generation requiring subjective evaluation, high for outcomes that pass/fail

deterministically (a structure's GDT score, a unit test, a CDT/ICD/X12 claim that adjudicates, a math answer that auto-verifies).

The filter is the conjunction. A vertical is a candidate data asset only if it scores high on BOTH axes. High-B-low-A verticals (code, competition math) are verifiable but their data is public, so the moat is thin and the gains may merely elicit latent base-model skill rather than teach new capability [related insight i07]. High-A-low-B verticals possess proprietary data but cannot generate a trainable reward signal, so RLVR cannot be applied cleanly. Only the high-high quadrant yields data that is both buildable into a reward signal and defensible against replication.

Scoring protocol. For each candidate vertical we (i) identify the canonical dataset and rate assembly difficulty on a 0-3 ordinal (0 = scrapable web, 1 = aggregatable with effort, 2 = requires licensing/partnership, 3 = requires privileged access or de-novo generation); (ii) identify the success criterion and rate verifiability 0-3 (0 = subjective, 1 = rubric-scored by humans, 2 = checkable with tooling, 3 = deterministic pass/fail); (iii) place the vertical in the resulting 4x4 grid. We treat the product $A \times B$ as the screening score, with a hard floor that both axes must be ≥ 2 .

We validate the filter retrospectively: every historical win in our corpus (Section 4) should land in the high-high quadrant, and the difficulty of each benchmark should track its axis scores (low-A verticals saturate fast once models scale; high-A verticals stay open). Where the retrospective fit holds, we use the same scoring forward in Section 7. We explicitly do NOT claim the filter measures willingness-to-pay, regulatory feasibility, or competitive timing — those are out of scope and addressed in Limitations (Section 8).

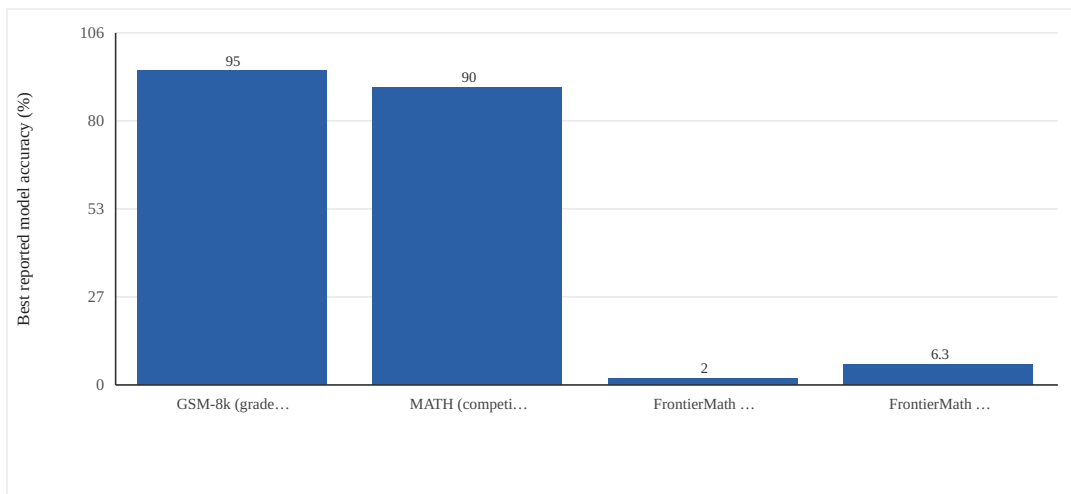


Figure 1. Verifiable yet unsolved: where headroom (and thus data value) remains. All four benchmarks are machine-verifiable, but difficulty differs by orders of magnitude. Near-perfect GSM-8k/MATH scores coexist with $<2\%$ on FrontierMath overall and 6.3% on Tier-4 (o4-mini; most leading models score 0% on Tier 4) [5, Epoch AI]. Verifiability alone does not make a benchmark easy — but FrontierMath's open mathematical corpus means its difficulty is not an owned asset. GSM-8k/MATH values are approximate frontier figures cited for contrast. Glazer et al. 2024 (FrontierMath) [5]; Epoch AI FrontierMath Tier-4 (o4-mini 6.3%). GSM-8k/MATH $\sim 95/\sim 90$ are well-documented approximate frontier scores. · Empirical data.

4. Evidence: The Template Across Four Unrelated Wins

Protein structure (high-A, high-B). AlphaFold2 was trained on experimentally-resolved structures from the Protein Data Bank and scored a median 92.4 GDT_TS at CASP14, with backbone accuracy of 0.96 Å RMSD versus 2.8 Å for the next-best method (and a summed-z-score margin of 244.0 vs 90.8 for the next group) [1]. Both axes are maximal: the PDB is the product of decades of irreproducible wet-lab crystallography ($A=3$), and GDT is a fully

automatic metric (B=3). ESM-2 then scaled the same structured-biology data to 15B parameters, with atomic-level structure emerging and 617M metagenomic structures predicted (772M by the March 2023 update) [2][3] — confirming the asset compounds.

Code (high-B, lower-A). SWE-bench draws real GitHub issues whose fixes are verified by tests; at launch the best system (Claude 2) resolved only 1.96% [4]. Verifiability is maximal (B=3, tests pass or fail), but the underlying data is public GitHub (A≈1). The filter predicts a thin moat, and the trajectory bears it out: rule-based RL on public software-evolution data alone took a sub-100B model to 41.0% on SWE-bench Verified [16], and the frontier has since crossed 80.9% [21] — a >40x rise on a public benchmark, with the capability diffusing across labs once models scaled. The defensibility gap shows up the moment the data stops being public: on SWE-Bench Pro, the same frontier models reach ~43.6% on the public set but only 17.8% on the held-out proprietary commercial repositories [17]. Public-data verticals deliver verifiability without defensibility.

Mathematics (high-B, lower-A). FrontierMath uses original, unpublished problems with automated verification; leading models solve less than 2% despite near-perfect GSM-8k and MATH scores [5]. The best Tier-4 result is o4-mini at 6.3%, with most leading models at 0% on Tier 4 [Epoch AI]. Verifiability is maximal and the frontier is genuinely hard, but the corpus is mathematical knowledge in the open — the moat is in the problem-authoring, not in proprietary data.

Finance (high-A, high-B). BloombergGPT's edge traced to a 363-billion-token proprietary financial corpus; it led 4 of 5 public financial tasks and scored 62.5% average versus 51.9% for GPT-NeoX, with a 25-30 point NER/NED improvement over GPT-NeoX and OPT-66B [6]. The corpus (A=3, four decades of licensed market data) is the moat; many financial outcomes (sentiment-vs-return, NER against ground truth, ConvFinQA numerical answers) are checkable (B≈2).

Table 1. Every win pairs a hard-to-assemble dataset with a machine-checkable outcome. Code and math (public data) deliver verifiability without defensibility.

Vertical / system	Proprietary data asset (axis A)	Verifiable outcome (axis B)	Headline result (verbatim-sourced)	Source
Protein structure / AlphaFold2	PDB experimentally-resolved structures (decades of crystallography)	GDT_TS automatic metric	Median 92.4 GDT_TS at CASP14; backbone 0.96 Å vs 2.8 Å next-best	[1]
Protein structure / ESM-2	Structured biological sequence data at scale	Structure recovery / accuracy	Up to 15B params; 617M (772M by March 2023) metagenomic structures predicted	[2][3]
Code / SWE-bench	Real GitHub issues (public, axis A≈1)	Unit tests pass/fail	Best system (Claude 2) resolved 1.96% at launch; frontier SWE-bench Verified since reached 80.9% [21], while SWE-Bench Pro's proprietary commercial set stays at 17.8% [17]	[4][17] [21]
Mathematics / FrontierMath	Original unpublished problems (knowledge open)	Automated answer verification	Leading models solve <2% (Tier-4 best o4-mini 6.3%)	[5]
Finance / BloombergGPT	363B-token proprietary financial corpus	Sentiment/NER/ConvFinQA checkable	Leads 4 of 5 financial tasks; 62.5% avg vs 51.9% GPT-NeoX	[6]

Jumper et al. 2021 [1]; Lin et al. 2023 [2]; ESM Atlas [3]; Jimenez et al. 2024 [4]; Glazer et al. 2024 [5]; Wu et al. 2023 [6]; SWE-Bench Pro [17]; Claude Opus 4.5 SWE-bench Verified 80.9% [21].

Table 2. BloombergGPT's proprietary-corpus edge: best of compared models on 4 of 5 tasks (NER essentially tied), with a 25-30 point NER/NED improvement over GPT-NeoX and OPT-66B on the broader evaluation. Per-task scores from Table 8 of the paper.

Task	BloombergGPT	GPT-NeoX (20B)	Notes
FPB (sentiment)	51.07	44.64	Financial Phrase Bank
FiQA SA	75.07	50.59	Financial sentiment analysis
Headline	82.20	73.22	News headline classification
NER	60.82	60.98	GPT-NeoX edges out by 0.16
ConvFinQA	43.41	30.06	Conversational financial QA
5-task average	62.51	51.90	BloombergGPT leads 4 of 5 tasks

Wu et al. 2023, BloombergGPT, arXiv:2303.17564, Table 8 [6]; per-task scores verified against the arXiv HTML.

Table 3. The 2025-2026 market is independently pricing both filter axes — paying a 4-5× premium for exclusivity and targeting regulated, checkable verticals. These figures evidence category demand and an exclusivity premium, not vertical-specific willingness-to-pay (see Limitations).

Signal	Reported figure	Relevance to filter	Source
Anthropic annual RL-environment spend (discussed)	>\$1 billion / year	Category-level demand	[8]
Typical environment contract	6-7 figures per quarter	Buyer budgets exist	[7]
Per-task pricing	\$200-\$2,000 (rare \$20k)	Unit economics of verifiable tasks	[7]
Compute per task (RL training)	~\$2,400	Cost structure of environments	[7]
Exclusivity premium	4-5× non-exclusive price	Market prices proprietariness (axis A)	[7]
Surge reported revenue	~\$1.2B (prior year)	Incumbent pivot to environments	[8]
Mercor valuation	~\$10B; pitches coding/healthcare/law	High-A/high-B verticals targeted by name	[8]

Epoch AI 'An FAQ on Reinforcement Learning Environments', Sept 2025 [7]; The Information via TechCrunch, Sept 21 2025 [8].

5. The Market Has Discovered the Template (2025-2026)

If the proprietary-data × verifiable-outcome template is the right abstraction, capital should be flowing toward exactly that structure now that the data wall is in view. It is. A new class of 'RL environment' companies — verifiable, often proprietary task simulators — emerged in 2025, and the labs are buying [7][8].

The figures are concrete. Anthropic leadership reportedly discussed spending over \$1 billion on RL environments over a single year [8]. Environment contracts run six-to-seven figures per quarter; per-task pricing sits at \$200-\$2,000 with rare \$20k outliers; a Slack-style product clone runs ~\$300k and a simple website replica ~\$20k, with roughly \$2,400 in compute spent per task during RL training [7]. Crucially for the defensibility axis, exclusive environment deals are priced roughly 4-5× higher than non-exclusive ones [7] — the market is literally pricing the proprietariness axis.

The data-labeling incumbents are pivoting toward this structure. Surge reportedly generated ~\$1.2B in revenue from work with major labs and spun up environment work; Mercor, valued at ~\$10B, is pitching investors on

building RL environments for domain-specific tasks 'such as coding, healthcare, and law' [8]. Prime Intellect is building a 'Hugging Face for RL environments,' and Mechanize is paying engineers ~\$500k to build them [8]. Healthcare and law — high-A, high-B verticals by our filter — appear by name in the pitches.

This is independent market confirmation, not of the paper's thesis being novel to the market, but of the template being the operative selection logic. The premium for exclusivity and the targeting of regulated, checkable verticals are precisely what the two-axis filter predicts the market should value.

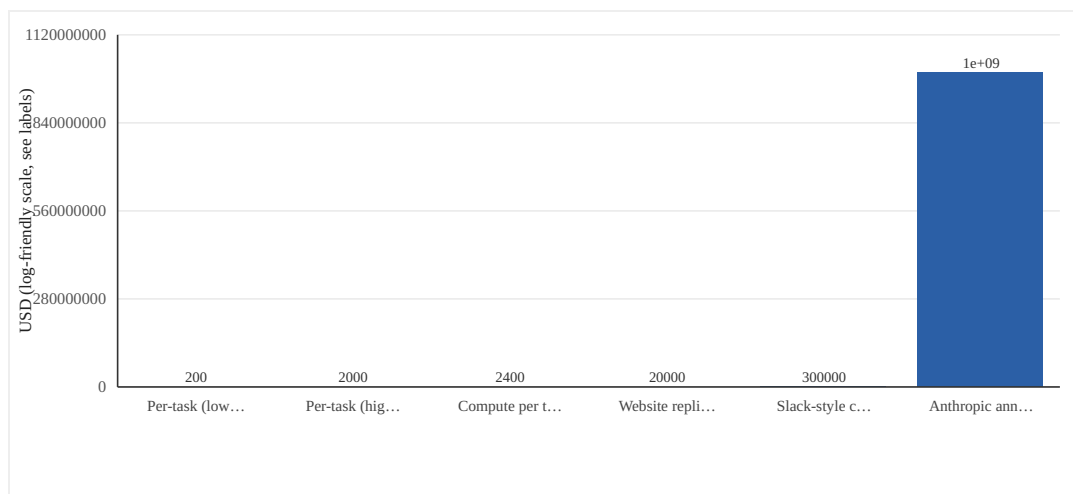


Figure 2. The market prices both axes: RL-environment economics, 2025-2026. Reported 2025-2026 pricing for verifiable RL environments. Per-task pricing runs \$200-\$2,000 (rare \$20k); ~\$2,400 compute per task in RL training; a website replica ~\$20k; a complex product clone ~\$300k; and Anthropic leadership reportedly discussed >\$1B in annual environment spend. Exclusive deals (the proprietariness axis) command 4-5× the non-exclusive price [7][8]. Wide range; figures are reported point estimates, not a continuous series. *Epoch AI 'An FAQ on Reinforcement Learning Environments' Sept 2025 [7]; The Information via TechCrunch (Anthropic >\$1B) [8]. · Empirical data.*

6. Analysis: Why Both Axes Are Necessary

The two axes are not redundant; each defends against a distinct failure mode. Without verifiability (low B), there is no automatic reward signal, so the data cannot be turned into RLVR training pressure and improvement depends on expensive, noisy human judgment that does not scale [10][12]. Without proprietariness (low A), the data is replicable, so any capability built on it is competed away — and, worse, the gains may be mere elicitation of latent base-model skill rather than genuine teaching, because public data is already in pretraining [related insight i07].

The benchmark trajectories make the asymmetry visible. SWE-bench (high-B, low-A) is checkable by construction, so progress is measurable and fast once models scale — from 1.96% at launch [4] to a frontier above 80% within two years [21] — but its low proprietariness means any capability built on public GitHub data diffuses to every lab. AlphaFold (high-A, high-B) did not 'saturate and diffuse' the same way, because the PDB-scale experimental asset is not replicable on demand — the moat persisted into ESM-2 and the metagenomic atlas [1][2].

FrontierMath illustrates a subtle case: a high-B vertical can stay unsolved for a long time (<2% [5]) without being defensible, because the difficulty lives in problem-authoring rather than in owning data. Open difficulty is not the same as an owned asset. This is why the filter requires both axes ≥ 2 rather than rewarding either alone.

The general principle: verifiability determines whether you can train on the data; proprietariness determines whether the resulting capability is yours to keep. A durable vertical data asset needs both — the trainability of code and the ownership of the PDB.

7. Implications for Data Strategy

Applying the filter forward nominates a shortlist of high-high verticals: clinical claims/coding adjudication (proprietary EHR/claims data paired with CDT/ICD/X12 codes that deterministically pass or fail), insurance adjudication, lab/diagnostic-test workflows, and contract/legal review (privileged corpora with checkable clause outcomes). Each pairs an asset that is expensive or privileged to assemble ($A \geq 2$) with an outcome a machine can adjudicate ($B \geq 2$). 2026 contamination-controlled benchmarks confirm these remain genuinely unsolved rather than already saturated: on the contamination-free LiveMedBench the best of 38 LLMs scores only 39.2% [20], and on FrontierFinance — 25 real financial-modeling tasks averaging 18+ expert-hours each — human experts still beat SOTA systems [22]. Open headroom on proprietary-data verticals is exactly where a defensible data asset earns its return.

Dental claims/coding is the worked high-fit example: a private clinical record (audio→exam→treatment plan) paired with a deterministic CDT-coding/claims outcome. It sits in the high-high quadrant on data shape alone — which is exactly what the filter is designed to surface. The market signal corroborates the direction: Mercor explicitly pitches healthcare RL environments, and exclusive deals command a 4-5× premium [7][8], rewarding the proprietary clinical-data anchor.

The strategic move follows the BloombergGPT and AlphaFold precedent: own or exclusively license the real-data anchor and treat the modeling method as a commodity. Methods (LoRA, GRPO, distillation) are published and copied within months; the corpus is not [related insight i11]. Capital and exclusivity should concentrate on the data, not the recipe.

Finally, the filter pairs naturally with a recurring product. Because verifiable held-out evaluations decay through contamination, the same proprietary-plus-verifiable asset supports both a training-data offering and a continuously-refreshed eval — a subscription-shaped data stream rather than a one-time dump [related insight i09].

8. Limitations and Counter-Evidence

Scope of the filter. This filter scores only two properties — data-proprietaryness and outcome-verifiability. It is explicitly necessary, not sufficient. It does not score AI-automation alpha, vertical monopoly dynamics, AI-integration greenfield, market-entry ease, or competitor data-vendor gaps. A vertical can pass the data-shape filter and still fail on regulatory feasibility (e.g. HIPAA constraints on clinical-data resale), supply (a single-clinic corpus of $n \approx 673$ records is small for true RL), or willingness-to-pay. The go/no-go decision must run the full lens, not this filter alone.

Demand is a multiplier, not a budget. The 2025-2026 market figures [7][8] evidence a structural demand surge and an exclusivity premium, but they do not establish absolute willingness-to-pay for any specific vertical. Anthropic's reported >\$1B and the 4-5× exclusivity multiple are real signals of category value, not a quote for a clinical-claims environment. Buyer budgets must be validated directly.

Counter-evidence on the verifiability axis. RLVR gains can be partly illusory: on Qwen2.5-Math-7B, RLVR improved MATH-500 even with random (+21.4 pts) or incorrect-label (+24.1 pts) rewards, nearly matching the +29.1 pts from ground-truth rewards, yet the same spurious rewards failed to help — and sometimes hurt — Llama3 and OLMo2 [related insight i07, 15]. This means a checkable outcome does not guarantee that training on it teaches new capability — it may only elicit latent skill (and is partly attributable to benchmark contamination in the base model). Contamination compounds the risk: on SWE-bench Verified, SOTA models locate the buggy file from the issue text alone with up to 76% accuracy in-benchmark but only 53% on out-of-benchmark repositories, evidence of

memorization rather than reasoning [18]; and on the contamination-free LiveMedBench, 84% of models degrade on post-cutoff cases [20]. The filter mitigates this by requiring high proprietariness (data the base model never saw), but the risk that verifiable signal in a public domain merely reveals rather than teaches is real and model-dependent.

Benchmark evidence and its limits. We anchor the SWE-bench evidence on the primary-source launch number (1.96% [4]). The post-launch curve, which earlier drafts left unquantified for lack of primary sources, is now anchored on peer-reviewed/arXiv reports rather than aggregator leaderboards: open rule-based RL on public software data reached 41.0% [16], and frontier SWE-bench Verified crossed 80.9% by late 2025 [21]. We still avoid stitching heterogeneous leaderboard submissions into a single controlled trajectory, and note that contamination-resistant successors keep the genuine frontier low — 17.8% on SWE-Bench Pro's proprietary commercial set [17]. The broader point is unchanged and now better evidenced: a high-verifiability, low-propietariness benchmark improves quickly and diffuses across labs, which is why such benchmarks make poor data moats.

9. Conclusion

AI's largest vertical wins are not a list of unrelated miracles. AlphaFold, SWE-bench-driven code progress, FrontierMath, and BloombergGPT share one structure: a hard-to-assemble dataset crossed with a machine-checkable outcome [1][4][5][6]. That structure is transferable. Cast as a two-axis filter — proprietariness × verifiability — it retrospectively fits every win in the historical record and prospectively nominates a shortlist of regulated, checkable verticals (clinical claims/coding, insurance, diagnostics, legal review).

The timing argument is external and strong. The human-text data wall arrives 2026-2032 [9], and the market has already begun pricing proprietary verifiable environments at a premium, with exclusive deals at 4-5× and lab budgets in the billions [7][8]. The scarce, defensible supply of verifiable signal is exactly the high-high quadrant the filter identifies.

We close where rigor demands: the filter scores data shape, not business viability. It is a fast, evidence-grounded way to nominate candidates and to reject the deceptively attractive high-verifiability/low-propietariness verticals (code, math) whose moats are thin. The full investment decision still requires the complete lens, real buyer validation, and a clear-eyed accounting of regulatory and supply constraints.

INVESTMENT THESIS

The pattern behind every durable AI vertical win is the same: proprietary/hard-to-assemble data × a machine-checkable outcome. Score targets on those two axes and only fund the high-high quadrant — the market already proves the thesis by paying a 4-5× premium for EXCLUSIVE verifiable environments and routing >\$1B/yr (Anthropic alone, per The Information) into the category, with Mercor (~\$10B) explicitly pitching healthcare and law. Code and math look attractive (perfectly verifiable) but score low on proprietariness, so their moats evaporate: SWE-bench launched at just 1.96% [4] yet, being built on public GitHub data, the frontier raced to 80.9% [21] and the capability diffused to every lab — while the moment the repos are held-out and proprietary, the same models drop to 17.8% (SWE-Bench Pro) [17]. The defensible bets are regulated, checkable verticals where the data is privileged: clinical claims/coding, insurance adjudication, diagnostics, legal review. CAVEAT: this filter scores data shape, not business viability — it nominates; it does not green-light. Validate willingness-to-pay, data-resale/HIPAA rights, and corpus supply before writing the check (a single-clinic n≈673 corpus is too thin for true RL).

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